Plant response to elevated carbon dioxide levels

an integrated survey of studies on the biochemical adaptations of different photosynthetic pathways and nutrient cycles of plants

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1. Introduction

Photosynthesizing plants dominate the ecosystems of the world today. They utilize the energy of sunlight to process carbon dioxide, water, oxygen and various nutrients, which they extract from the environment, to synthesize the elements needed for their growth and maintenance. When one of these compouonds is in low supply, the growth of the plant is limited. The reverse is also true: if a certain compound is in abundant supply it will not only stop being a limit to plant development but through interdependancy of resources utilization processes it can have a stimulating effect in general.
Atmospheric levels of carbon dioxide have risen from 280 to 380 ppm over the last two centuries (Takahashi, T.; 2008) and are expected to keep rising. This essay attempts to provide an insight in the possible implications for plants and thus ecosystems worldwide. To properly adress the issue first several processes that regulate plant growth and development must be considered, and their interdependancy with carbon dioxide levels. Then an integrated survey of current knowledge of the effects on plant biology and development will be presented.

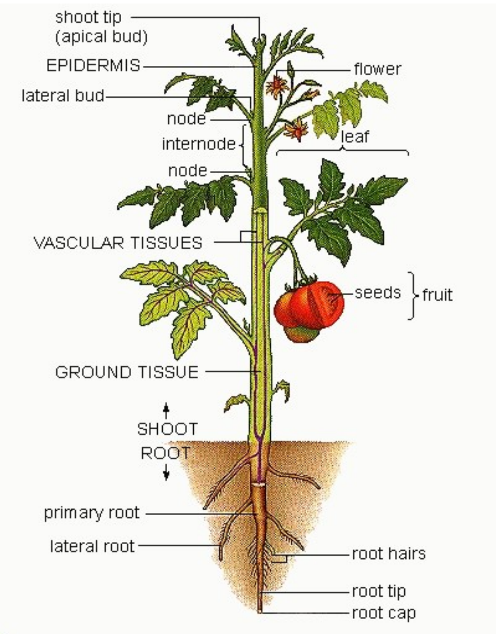
2. Plants

2.1 Introduction

The organisms on earth that consist of multiple cells, have the ability to produce their own food through photosynthesis and are adapted to life on the land surface are called autotrophic multicellular eukaryotes, or plants, and belong to the classification Kingdom Plantae. This Kingdom is divided in vascular (tracheophytes) and non-vascular plants (bryophytes), that use conducting tissues to transport resources within their forms. The vascular plants also distinguish themselves by their ability to lignify their cell walls, resulting in greater stability and thus greater plant size; the non-vascular plants are generally smaller. The vascular plants are further divided into seedless- and seed plants and the latter of these two types has dominated the flora world for several hundred million years due to the advantages provided by a combination of photosynthesis, efficiency, size and seed formation, with improved survivability and thus ensured reproduction.
The vascular seed plants are further divided into gymnosperm or naked-seed plants that carry their seeds on their leaves, and angiosperms or flowering plants that wreathe their reproductive organs in flowers. These latter plants are the most common and widespread kind of plant and unless other wise is specifically stated most arguments will be based on the angiosperm plant form.

2.2 Physiology

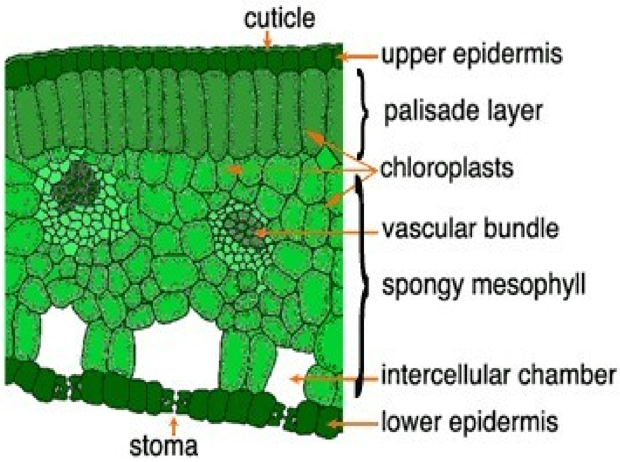
As a reference the organization of the plant body will be summarily described here. Basic physiology can be derived from the image to the right. The plant draws water and nutrients out of the soil through its roots and transports this to the photosynthesizing parts of the body, which commonly means the leaves, where the water is used in synthesizing food for the plant yet over 90% of all water is lost to evaporation. With these photosynthetic centers the plant also draws oxygen and carbon dioxide in from the atmosphere, releases the same two compounds from different processes, and strives to achieve an optimal light capture position through growth and movement. For reproduction the plant develops flowers to spread its pollen and/or receive it, commonly attracting agents such as insects to facilitate the transport. After fertilization a fruit is developed which is released and transported through various means to germinate and possibly grow to a mature specimen itself.



2.3 Tissues

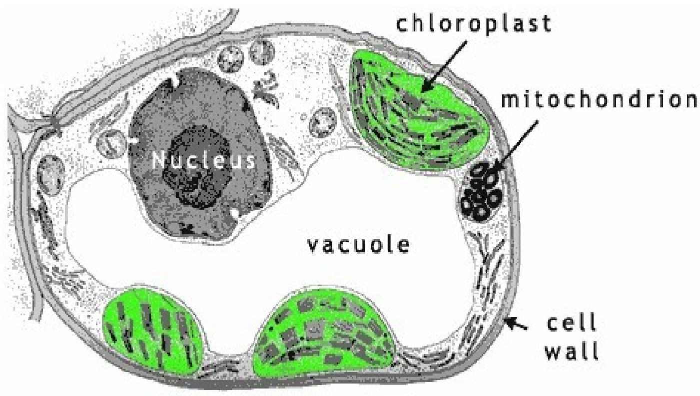
In more detail several systems within the plants tissues can be described. The vascular system is composed of two main tissues, the xylem and phloem. Both are networks of transporting cells and where the xylem is responsible for the transportation of water and thus dissolved nutrients, the phloem is mostly used for the transportation of synthesized products of photosynthesis. Although not entirely exclusive to the named compounds, the water transport predominance of xylem requires it to be larger then phloem. Filling the space between the vascular system and the harder epidermis or outer skin of the plant are found several tissues that provide physical support, storage and other functions such as photosynthesis, regeneration and healing of wounds, respiration and more.

All these tissues are also found within the leaves. In the inner tissue or mesophyll we find the parenchyma cells which are specialized for photosynthesis, consisting in the example here of spongy parenchyma and palisade parenchyma cells. Within the vascular bundles the xylem cells transport water and minerals from the soil to the cells, while the phloem cells carry off the products of photosynthesis to various parts of the plant. Oxygen, carbon dioxide and other gasses enter and exit the intercellular spaces of the leaf through many small openings located in the epidermis of the leaf called stomata. Their aperture can be regulated by the plant through several mechanisms that respond to carbon dioxide levels in the intercellular air, water stress and temperature to reach an optimal balance between those factors and ensure a healthy photosynthetic activity without undue water loss under unfavorable conditions.



2.4 The Cell

Although cells are incredibly varied in their structure and functions, we shall take a closer look at a simplified example to discuss the common basic of cell anatomy. From the simplified illustration below we can see that the cell consists of several distinct parts. The cell is encased in a cell wall that allows transport of substances through it, and is filled with liquid cytoplasm and a nucleus, the control center of the cell that holds all genetic information. Other components of note and some of their typical functions are the chloroplasts, responsible for photosynthesis, the mitochondria, responsible for respiration and the vacuole which is vital for storage and structural support. To take a further look into the these cellular components would be fruitless without simultaneously looking at the functions they perform, so we will continue the physiological analysis into the description of photosynthesis and respiration.



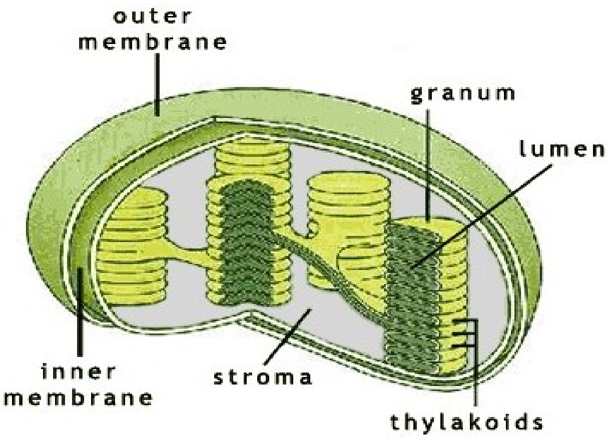
3. Photosynthesis and respiration

3.1 Introduction

According to the second law of thermodynamics, if no energy enters or leaves a system the potential energy of the final state will always be less then that of the initial state. This also means that any reaction lacking an input of energy will always have final reactants with a lower potential energy then its initial reactants with the difference lost in the reaction itself as heat, light or other forms. From this it can be derived that if no energy enters or leaves a plant, its metabolism will inevitably let the plant dwindle into a increasingly low energy state and die. To ensure their survival, plants release waste products much like other life forms but are unique in their method of taking in energy. They are able to harness the energy of sunlight and store it in the bonds of synthesized complex molecules which are transported throughout the plant to provide the neccesary inputs of energy. This process is called photosynthesis and it is the way by which almost all energy enters our biosphere. As a chemical process it is largely dependant upon abiotic factors and resources such as carbon dioxide, water, oxygen, nutrients and temperature and it occurs in several forms in different species of plants, named the C3, C4 and CAM pathways, that each have distinct advantages or disadvantages to eachother depending on their environment and that make them ideally suited for specific conditions, mostly due to their different use of and dependancy on aforementioned resources.

3.2 Chloroplasts

As will be remember from the previous information on plant physiology, it is the chloroplasts that are responsible for the photosynthetic process. The parenchyma cells within the leaf are exposed for much of their surface to the intercellular space which provides the chloroplasts access to the gasses flowing through the leaf. A closer look at a chloroplast yields the following image:



The chloroplast consists of an outer and inner membrane, stacks of disklike thylakoids called grana and stroma thylakoids which connect the grana through the stroma or inner space of the chloroplast. It is in the thylakoid membranes that the actual photosynthesis process takes place.

3.3 Photosystems I and II

Here two distinct mechanisms are at work, photosystems I and II. Some plants are not equipped with photosystem II, which will be of some importance later on in the discussion of abiotic factors. In photosystem II light is absorbed by pigment molecules called chlorophylli either directly, or indirectly through antenna molecules. An electron of the molecule is excited, which means raised to a higher energy state, by this light energy and passed on along an electron transport chain to photosystem I. The chlorophyll draws another, low-energy electron from a water molecule, thereby photolysing the water molecules streaming through the tylakoids into protons and oxygen gas. While the oxygen gas is transported out into the stroma and eventually out of the chloroplast and into the intercellular space, the protons are released into the thylakoid lumen or inner space, from which they are pumped to the stroma and out of the choloroplasts. It is this electrochemical gradient from the proton-rich thylakoid lumen to the proton-poor stroma that powers ATP synthesis in the chloroplast, named photophosphorylation and discussed in the next paragraph.
The electron that was excited in photosystem II arrives in photosystem I where it is accepted and once more activated by another pigment molecule that functions in a similar way. The excited electron is moved down a chain of carriers to reduce the coenzyme electron carrier $NADP^+$ to NADPH and the pigment molecule draws another electron from photosystem II. In formulae:

Photosystem II:
$H_2O \rightarrow 2\ e^- + 2\ H^+ + 0.5\ O_2$ (light driven) P.1

Photosystem I:
$NADP^+ + H^+ + 2\ e^- \rightarrow NADPH$ (light driven, reversible to yield energy) P.2

Combined photosystems:
$H_2O + NADP^+ + H^+ \rightarrow 2\ H^+ + 0.5\ O_2 + NADPH$ P.3

When electrons are transported from water through photosystems I and II to the Calvin cycle discussed below, this unidirectional flow is called non-cyclic electron flow. Photosystem I can also work by itself, where the excited electron passes its energy down a chain of carriers and returns to the the core of photosystem I to be excited again, which is called non-cyclic flow.
The two products of NADPH and ATP are the most common forms in which the captured light energy is transmitted to power processes elsewhere within the plant.

3.4 Interlude: ATP
To drive endergonic or energy-requiring reactions, plants couple such reactions with exergonic or energy-releasing reactions. The medium for this energy tranport is most commonly the compound known as ATP or adenosine triphosphate. When a phosphate group is removed from this molecule by hydrolysis, a relatively high amount of energy is released, converting the ATP into ADP (D for di-) and a phosphate group.When repeated this ADP yields AMP and another phosphate group, while AMP can be 'recharged' to ADP and that to ATP through phosphorylation, the addition of a phosphate group. The energy release of this initial reaction is used in other reaction that require energy to function, such as the Calvin Cycle of the next paragraph. Note that when a phosphate group is removed from ATP it is usually added to another molecule, giving that process the same name of phosphorylation which should not be confused with ATP regeneration.

Phosporylation:
$ADP + Pi \rightarrow ATP + H_2O$ (proton gradient driven, reversible to yield energy) P.3

Where Pi is the phosphate group detached from ATP by oxidation. With the input of ATP, endergonic reactions can actually become exergonic and thus proceed spontaneously, such as the formation of sucrose. Since it is the principal energy conveyor within the plant, it was allowed a paragraph of its own.

3.5 The Calvin cycle
The central mechanism to all photosynthesizing pathways is the Calvin cycle, named after scientist Melvin Calvin. In it, carbon dioxide is combined to RuBP-molecules, five-carbon sugars with two phosphate groups that 'circle' the Calvin cycle: they pass through a looping series of reactions. After the mentioned molecules combine into three 6-carbon molecules they are quickly broken down to six 3-carbon PGA-molecules. These are in turn reduced with NADPH to form six 3-carbon PGAL molecules, and of these five are reducted with ATP to form RuBP again and resume the cycle from the beginning, with a net gain of one PGAL-molecule per three cycles; since one carbon dioxide molecule is added per turn and PGAL consists of three carbons it takes three cycles to fully synthesize one PGAL and export it to the cytosol where it is converted to sucrose. To summarize the cycle in a formula:

$3\ CO_2 + 9\ ATP + 6\ NADPH + 6\ H^+ \rightarrow PGAL + 9\ ADP + 8\ Pi + 6\ NADP^+ + 3\ H_2O$ C.1

The ADP, Pi and $NADP^+$ return to photosystem I where they are converted to ATP and NADPH.

The PGAL is transferred to the cytosol of the cell where it can be converted to glucose. In a simplified form this can be formulated as such:

2 PGAL + 2 ADP $\rightarrow$ Glucose + 2 ATP C.2

Combined with formula P.3 this gives:

2 PGAL + 2 H_2O $\rightarrow$ Glucose + 2 Pi (and vice versa) C.3

However, very little PGAL is actually converted to glucose right away. Rather, it is converted to sucrose, the primary transport sugar of the plant, or starch, the primary carbohydrate storage compound. During respiration it is converted back to PGAL in the process of glycolysis. These processes are identical, although reversed, and since the total formula scheme remains accurate with only a textual description of these processes they have been omitted for the sake of clarity.

3.6 Mitochondrial respiration

The products of photosynthesis are transported throughout the plant and stored to be used when needed for the release of energy. The process responsible for almost all of the energy release from these compounds is called respiration which takes place in the mitochondria and is defined as the complete oxidization of sugars and other organic molecules to carbon dioxide and water.
First the starch or sucrose stored in the cell is converted to glucose or sucrose through a reverse of C.3. Then it can enter the process of respiration proper starting with glycolysis where it is broken down to form pyruvate, drive ATP synthesis and reduce the electron carrier NAD^+:

Glucose + 2 NAD^+ + 2 ADP + 2 Pi $\rightarrow$ 2 Pyruvate + 2 NADH + 2 H^+ + 2 ATP + 2 H_2O R.1

Then the pyruvate is combined with the facilitating enzyme CoA to form acetyl CoA and reduce more NAD^+:

Pyruvate + NAD^+ + CoA $\rightarrow$ CO_2 + Acetyl CoA + NADH + H^+ R.2

It is in this form of acetyl CoA that the carbon atoms from glucose enter the citric acid cycle. In this cycle a series of oxidization reactions consumes the acetyl CoA and yields carbon dioxide, drives more ATP synthesis and reduces the two electron carriers NAD^+ and FAD.

Oxaloacetate + Acetyl CoA + 3 H_2O + ADP + Pi + 3 NAD^+ + FAD $\rightarrow$ Oxaloacetate + 2 CO_2 + CoA + ATP + 3 NADH + 3 H^+ + $FADH_2$ R.3

NADH and $FADH_2$ now carry highly energetic electrons that proceed to move down a electron carrier tranport chain, releasing free energy with each step. This reducts and subsequently oxidizes the carriers powering a proton gradient along the chain and thereby facilitates the phosphorylation of ADP to form ATP, and ultimately reduces and combines H^+ and O2 to water. 3 ATP molecules are synthesized by NADH oxidization, whilst $FADH_2$ yields 2.

2 H^+ + O_2 + NADH $\rightarrow$ H_2O + NAD^+ and 3 ADP + 3 Pi $\rightarrow$ 3 ATP + 3 H_2O
2 H^+ + O_2 + $FADH_2$ $\rightarrow$ H_2O + FAD and 2 ADP + 2 Pi $\rightarrow$ 2 ATP + 2 H_2O

Combined:
3 ADP + 3 Pi + 2 H^+ + O_2 + NADH $\rightarrow$ 3 ATP + 4 H_2O + NAD^+ R.4
2 ADP + 2 Pi + 2 H^+ + O_2 + $FADH_2$ $\rightarrow$ 2 ATP + 3 H_2O + FAD R.5

The overall ratio of this process is as follows:

$$1\ C_6H_{12}O_6\ \text{(glucose)} + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + 36\ ATP \qquad \text{R.6}$$

It is interesting to note that under stressfull conditions other forms of the cycle can exist. If there is not enough oxygen available, glucose can be converted by anaerobic processes called fermentation to yield a severely diminished 2 ATP molecules. It can also convert fats and proteins to molecules that can enter the respiratory sequence at various steps, with various penalties as to efficiency.

3.7 'Dark' respiration
RuBP is not very discriminate towards carbon dioxide and oxygen. If in ample supply, it can combine with oxygen resulting in a very different process involving the chloroplast, peroxisome organelles and mitochondria which is quite unproductive; it doesn't yield ADP of NADPH and consumes O_2 to produce CO_2. This process is called photorespiration or 'dark respiration' to distinguish it from mitochondrial respiration and might be an evolutionary relic from times when oxygen wasn't a major constituent of our atmosphere. As much as 50% of fixed carbon may be lost in this process which competes with photosynthesis in the plant. In several cases this 'dark' respiration can dominate over the photosynthetic process, the most obvious occurences are when the stomata connecting the inner leaf spaces to the outside air need to close because of water stress or related causes, or when the stillness of the air around the leaf leads to a shortage of carbon dioxide and rising oxygen levels from respiration proper.

3.8 C3, C4 and CAM pathways
Because the PGA molecule consists of three carbon atoms, the Calvin Cycle is also named the C3 pathway and plant that use only this pathway are classified as C3-plants.
However, in some plants the Calvin cycle is relocated from the mesophyll cells to the bundle sheath cells and are provided with carbon dioxide through a different process called the C4-pathway. In this process, carbon dioxide is combined in a catalyzed reaction with a compound named PEP to form oxaloacetate, a four-carbon compound which gives the pathway its name. This oxaloacetate is then reduced to malate or converted to aspartate, which is transported to the bundle-sheath cells where they it is carboxylated to yield carbon dioxide and pyrovate, the latter of which returns to the mesophylli to react with ATP to form PEP once more. The carbon dioxide enters the Calvin cycle proper. Because the fixation of carbon dioxide to PEP is not affected by the presence of oxygen, this 'conveyor belt' of carbon dioxide provides it a solution to the problem of respiration and a serious advantage over C3 pathways. However, because C4 plants need two additional ATP-molecules to convert pyruvate back to PEP they need five ATP to fix one molecule of carbon dioxide, as opposed to three for C3 plants. This energetic disadvantage is offset by mentioned greater efficiency in using the available carbon dioxide which results in C4 plants dominating in environments with high light-intensities, high temperatures and little water. However, due to the energetic disadvantage, C4 plants have trouble competing with C3 plants that require less energy to photosynthesize in low light intensities and temperatures below around 25º Celsius. The dominance balance between C3 and C4 is therefore mostly an issue of available carbon dioxide, light intensity, temperature and effective use of resources, as will be discussed in more detail later.

Another strategy for carbon dioxide fixation is that of crassulacean acid metabolism, or CAM. They use both the Calvin cycle and C4 pathway but have a temporal separation between the carbon fixation and sugar synthesis rather then a spatial one. In a way rather similar to that of C4 plants, CAM plants fix carbon dioxide at night, and form malate which is stored in the vacuole of the cell. When day comes, the plants stomata close and the stored malate is converted back to carbon dioxide which is put through the Calvin cycle to form sugars. The advantage of this mechanism is a greatly increased preservation of moisture, allowing the plant to live in very arid conditions. During prolonged drought, the plant can even keep its stomata closed at night and refix the carbon dioxide produced by photorespiration. Offsets of these advantages are a slow growth, which reduces their competitiveness with C3 and C4 plants, but in their designated habitats they are undisputably the dominant plant type.

These are the basics of the photosynthetic processes that commonly occur in plants. For the balancing of all these formula to ensure their correctness, see appendix A.
Now some details of required nutrients, specifically nitrogen, will be given.

4. Nutrients

4.1 Introduction

Plants require water, light and carbon dioxide to provide for their energy requirements. They can also synthesize everything else they need, like amino acids and vitamins, with the help of inorganic nutrients from the soil. These nutrient requirements are reflected in the atomic composition of the plant which varies somewhat between species, age and location. Typically however there are 9 macro- and 8 micro-nutrients, so called due to their relatively high and low concentrations, that are deemed vital to all vascular plants. Two tables of these nutrients is listed to the right, organized from the smallest concentration to the highest. Each compound fulfills a specific function within the plant, with accompanying functional impairments if in insufficient supply.

Micronutrients
Molybdenum
Nickel
Copper
Zinc
Manganese
Boron
Iron
Chlorine

Macronutrients
Sulfur
Phosporus
Magnesium
Calcium
Potassium
Nitrogen
Oxygen
Carbon
Hydrogen

4.2 Soil acidity

The inorganic nutrients are obtained from the soil where they are mostly present as ions in the soil water. Positively charged ions are generally retained better then those with a negative charge since humus and clay particles typically have an excess of negative charge on their surfaces with which they can bind nutrients and protect them from the leaching effects of percolating water. Important to note here is that in general the overal pH of the soil and specifically the acidity derived from the carbon dioxide present in air-filled pores and emitted from respiring roots, which dissolves in the soil water to form bicarbonate acid and hydrogen, have an effect on the total of available nutrients: the protons can replace nutrients on the clay- and humus surfaces and lower the ion-binding capacity of the soil.

4.3 Biogeochemical cycles

The nutrients mentioned are obtained from the soil and after the plant dies and decomposes they are returned to it as a part of the greater biogeochemical cycles of these elements, the pathways they take throughout the earth's systems. Many of these are said to be 'leaky', they loose amounts to reservoirs from which plants cannot recover them such as ocean basins or specific volatile forms in the atmosphere which the plant cannot fixate. A very important example of such a cycle is that of nitrogen. The fixation and use of nitrogen is on equal footing with photosynthesis and respiration in regard to the importance for our biosphere; in fact, the most common deficiency impairing plant growth and development is that of nitrogen. Since the cycle is likely to be effected by carbon dioxide levels, it shall be put forward in more detail.

4.4 The Nitrogen-cycle

The nitrogen-cycle in regard to plants loses quantities, just as many other cycles, to soil erosion, runoff into oceans, harvests and fires that remove nitrogen to ocean basins and the atmosphere from which plants cannot obtain them. The cycle is replenished through several mechanisms, amongst which are the weathering of rocks that contain nitrogen and the industrial production of fertilizer. However, since weathering is a very slow process, and industrial fertilizer has only existed since 1914, without furter ways of obtaining nitrogen plants would not have enough to fulfill their needs. Nitrogen is also added to the soil through dead organic matter, which is decomposed and ammonified by soil-dwelling bacteria to yield ammonium, which is in turn used by the bacteria in aerobic conditions to provide energy for their fixation of carbon dioxide and thus growth; this process yields both energy for them and nitrate for plant, their major source of nitrogen. Another way for nitrogen to enter the soil is through nitrogen-fixation, where atmospheric nitrogen is directly fixed by certain bacteria. This is a very energy-demanding process and can be carried out by independent bacteria or bacteria in a symbiotic relationship with plants, where the plant provides energy and carbon and receives nitrate. For this energy, respiration is needed which releases oxygen; if this oxygen content rises too high however, an anaerobic process called denitrification is stimulated which converts nitrate to volatile compounds that are released to the atmosphere once more. A very high oxygen level can even be an irreversible inhibitor to the symbiotic nitrogen-fixing relationship. Nevertheless, it is this process which is responsible for most of the nitrogen that is fixed worldwide. Plants with highly productive bacteria symbiosis are even grown and left to decompose on fields that have yielded crops to harvest to increase nitrogen content of the soil as an alternative to fertilizer. So many of the processes involved in the nitrogen-cycle are directly or indirectly influenced by carbon dioxide levels. What this means to the plants behaviour shall be discussed in the following chapter.

5. Elevated carbon dioxide and its effects

5.1 Introduction

Globally, carbon dioxide levels have been rising and are expected to keep on doing so. In response to this change a great multitude of possible effects rears its head, and so far science is struggling to provide clear answers as to how our world will adapt. A cautious hope that biosphere responses will be able to buffer and reduce global CO_2 increases is maintained, as the idea of increased forest-covers and agricultural efficiency. As far as this pertains to plant biology, much interesting progress has been made which will be provided in an integrated form in this chapter.

As described in the sections on photosynthesis, respiration and nitrogen-cycling, carbon dioxide is a vital compound in many aspects of the life of plants. To understand the impact of elevated CO_2 levels, hereafter to be designated with eCO_2 as opposed to aCO_2 or ambient carbon dioxide levels, first the changes in the mechanisms that rely on CO_2 must be understood. Then the general view that biosphere production will rise as a result from eCO_2, the 'carbon dioxide fertilization effect', can be properly put forward.

5.2 Effects on C3 and C4 pathways

In the C3 pathway all carbon fixation is performed by Rubisco which has the problem of being somewhat indiscriminate to CO_2 and oxygen, resulting in occasional oxylase and 'dark' respiration where, in the presence of low carbon dioxide, the enzyme binds O_2 instead of CO_2. Under eCO_2, this process occurs less and more $CO_2$is fixated, which has led C3 plants to display increased growth rates and fruit production in a multitude of species such as orange trees (Culotta, E.; 1995). With a mention of diversity of reactions amongst species it is true that quite a few C3 plants display a net increase in primary productivity. However, many studies also found that this initial growth burst eventually reaches a new balance where the plants, although larger and more efficient, no longer grow with such an improved expedience. To understand why this occurs we must first look at three factors that can limit the net rate of CO_2 assimilation and thus vitality of a C3 plant. First, the Calvin cycle can be inhibited by the mentioned process of Rubisco oxylase. Second, if CO_2 levels are high, a shortage of RuBP regeneration will form the limiting factor. This can be caused by a low light intensity and thus shortage of NADPH and ATP to power the cycle, or other factors such as a shortage in other enzymes necessary for proper functioning. A third limitation occurs when the chloroplasts have a higher capacity to produce then the leaf can use; it imposed a maximum on the photosynthesis rate which is not affected by rising carbon dioxide or oxygen levels. This is called TPU-limitation or triose phosphate-limitation, named after the compound that plays a major role in

the chemical mechanism. Determining which process is the dominant limiter is a complex procedure but a certain delineations can be put offered (Sharkey, T. D. et al.; 2007). Generally, the major limiter is accepted to be RuBP regeneration when the carbon dioxide levels are above 300 $\mu Pa\ Pa^{-1}$, and Rubisco-oxylase when below 200 $\mu Pa\ Pa^{-1}$; UTP-limitation doesn't seem to have such a definable range of dominance. Between 20 and 30 Pa a transition zone exists where the focus shifts from one limiting process to the other.
This implies that when carbon doxide levels are above 300 $\mu Pa\ Pa^{-1}$, or 300 ppm, Rubisco is no longer the limiting factor and the C3 plant will instead be limited by a RuBP -shortage; adaptations to reduce this inhibiting shortage can include redirecting resources to optimize usage of light and water by extending its leaf and root systems and by closing stomatal openings.

As will be remembered, stomata are opened to let air flow into the leaves and closed to reduce water loss under very dry or hot conditions. A balance is maintained between CO_2 demands and water loss risk which at eCO_2 can shift towards smaller stomatal openings, and in new leaves and offspring even towards less stomata per leaf surface (Konrad, W. et al.; 2008). This is one of the explanations put forward to explain the 'levelling out' of growth increases under eCO_2; the plant effectively reduces its CO_2 access to previous levels through less air exchange with the atmosphere and thereby structurally reduces its evapotranspiration (ET). A study from 2007 showed that, expectedly, C3 plant species responded to eCO_2 better in terms of net ecosystem carbon dioxide exchange (NEE) then C4 species, yet it was the C4 species that were most effective in reducing evapotranspiration (ET) (Polley, H.W. et al.; 2007) . Nuances to the unprecise and still experimental nature of the study and heterogeneous responses between species as well as photosynthetic pathways must be made, with many results just barely being significant, yet the observations do suggest a general pattern.
Logically, from the knowledge on photosynthesis pathways we can expect these results. Where carbon dioxide is not overabundant, an increase in its concentration will result initially in increased NEE; this will be most profound in C3 species that can reduce the disadvantage of their Rubisco-based carbon dioxide fixation. After a certain increase in carbon dioxide intake however, the plant decreases its stomatal openings to preserve water loss out of genetic response to the higher CO_2 or because its balance between various functions that can limit growth has shifted towards other priorities. As can be seen in the root/shoot ratio, water/CO_2 stomatal response and even seasonal deciduous leaf regeneration, a plants metabolism will focus on optimising those processes which form the biggest impediment to its growth and survival [x]. This includes but is not limited to the need for water uptake as shown in this study; other optimisations may include better uptake and use of nutrients and the capturing of more light. As the most efficient way to make use of the new circumstances, the exact zone of acclimatisation to higher carbon dioxide will depend largely on the specific species. That said, expectedly C4 species, who are not afflicted by Rubisco's oxygen confusion, have been observed to switch to reducing water loss earlier in this process and reach their 'optimal' state with greater improvements there or in other optimizations at the cost of lower NEE increase.

5.3 Effects on CAM pathways

CAM plants have untill recently been under debate due to very conflicting studies resulting from great interspecies heterogeneity of eCO_2 response and complex metabolic mechanisms pertaining to this subject. However, from a general chemical assessment and accompanying studies some conclusion can nevertheless be drawn.
CAM plants regulate their day/night metabolic processes through several mechanisms, one of which is the carbon partial pressure within the leaf. As has been discussed. At night the leaves use PEPC to fix carbon dioxide and store it as malate inside the leaf. When the light begins to shine the stored malate is decarboxylated for use in the Calvin Cycle and the leaf shifts from PEPC-fixation to Rubisco-fixation, with the elevated intercellular partial CO_2 pressure from malate carboxylation triggering the closing of stomata. During the day the plant processes the stored malate and converts it to sugars untill the stored malate has reached low levels and the reduction in pressure opens the

stomata and heralds a similar transition phase as at dawn, with the difference that here Rubisco-fixation dominates over PEPC-fixation.
Some studies have shown increased increased carbon fixation rate at night, for instance in Agave Vilmoriniana (Szarek, S. R. et al.; 1987) by more effective PEPC-fixation. More improvements were observed in the transition phases when Rubisco-fixation was active and the CAM-plant could apply the eCO_2 C3-advantage. Specimens of Aechmea 'Maya' have been observed to produce increased carbon fixation at dawn and dusk (Ceusters, J. et al.; 2008), and although the plant under study showed an inability to export this extra carbon to use for growth or respiration effectively cancelling all effect on biomass production, other species have shown to be able to effectively do so nevertheless (Nobel and Israel, 1994; Nobel, 1996; etc.). Another effect was the increased water use efficiency or WUE in a similar fashion to other plants: eCO_2 led to smaller stomatal openings and a resulting increase in WUE of up to 50% was observed (Szarek, S. R. et al.; 1987). For plants that thrive specifically in arid regions this could be a major advantage and have repercussions both in ecosystem composition and possibilities for agriculture.

5.4 Nitrogen
The major limiting nutrient for plants, nitrogen, is dependent on carbon dioxide levels for parts of its cycling. Soil microbes oxidate ammonia to form nitrate and oxygen and fixate CO_2 with the released energy; the process of denitrification where microbes convert the nitrate to volatile compounds occurs only in the presence of very low oxygen concentrations. As plants have access to more carbon dioxide and establish a higher net carbon exchange rate, the carbon dioxide emitted from the roots will also increase (Culotta, E.; 1995). This will in turn stimulate both the process of nitrification by soil bacteria and their populations. Combined with extended root systems from increased growth, the improved availability of nitrogen to plants could meet the increased demand for nitrogen. However, several other mechanisms have an opposite effect and are able to counter this advantage. The increased carbon dioxide content of the soil will increase acidity which will reduced the availability of nutrients ions adsorbed to clay and organic particles. Increased nitrification will lower oxygen content of the soil which could have a stimulating effect on denitrification; the extra produced nitrate might be used by the grown soil microbe population itself (Culotta, E.; 1995); and with the increased growth from eCO_2, nitrogen content of leaves might decrease and adversely effect their decomposition rate. A study into these decomposition rates in relationship to eCO_2 did observe fluxes in nitrogen-immobilization from decomposition and -denitrification, yet no significant relationship to eCO_2 could be found (Ross, J. et al.; 2002). Another study into nitrogen-content of leaves as related to eCO_2 and even nitrogen availability itself necessarily postpones its conclusion until after further study (David McGuire, A. et al.; 1995). All these subjects effect the nitrogen-cycle and could prove to have a major impact on the global response to eCO_2; proof for all of them has been found yet an unambiguous answer to which will dominate under what condition has not been found [x].

5.5 Transgenerational effects
All the studies quoted so far have focused on relatively short-term effects. However, important long-term effects must be considered also. Two of these are maternal effects or the changing resource allocation to seeds as a result of the metabolic changes in the mother plant; and evolutionary aspects of populations adapting to the new situation. As a recent study implies, although both enhancing and diminishing effects on short-term results can be expected from multiple generations being exposing to eCO_2, in general the initial growth increase, if not balanced in time within the generation itself, will be inhibited over time with some species showing only half the response to eCO_2 over 5 generations of acclimatisation (Lau, Jennifer A. et al.; 2008). On the subject of evolution, other studies have so far only showed only weak to non-existent evolutionary responses to eCO_2 (Bazzaz et al. 1995; Steinger et al. 2007), but the possibility of evolutionary trends towards species with a consistent higher growth rate still exists.

6. Conclusion

All this information and study implies several things. Firstly, since most agricultural crops are C3 plants, if the increased productivity of those species proves to be sustained and finds a balance with the nitrogen-cycling of the soil, good prospects for global food production lie ahead. Rice crops in subhumid tropical climates have already been shown to produce higher yields under eCO_2 (De Costa, W.A.J.M. et al.; 2006) C3 crops have already been seen outcompeting C4 weeds, reducing the needs for pesticide and further increasing possibilities for agriculture. Also the possible reduced need for fertilizer and crop rotations provides advantages both directly in costs and yields but also indirectly; since the industrial nitrogen fertilizer produced yearly amounts to almost half the planets biological nitrogen fixation, and it is produced at high energy cost which is provided by fossil fuels, reduced need for fertilizer will reduce the amount of carbon dioxide released into the atmosphere. Some publications even estimates, quite optimistically, almost 10% of all agricultural increase in production over the period of 1985 to 1995 to be attributable to the increase in global carbon dioxide levels.

As stated, CO_2 is not the only compound that determines a plants development. Almost all internal processes of a plant are interdependent and as such, with increased growth an increased need for water, light and nutrients, specifically nitrogen, can be expected. Current insight into the effects of eCO_2 on nitrogen-cycling is not able to produce clarity yet. Also, with maternal effects further curbing the growth increases, the hopes that increased biomass production might act to reduce global carbon dioxide increase will have to be adjusted to expect only an increased buffer of larger plants.

But one thing seems to follow quite clearly from all studies involved: water use efficiency is improved for nearly all species and variants, which will have effects of our global biosphere to extends well beyond the scope of this essay. Expanding habitats, shifting ecosystem dominances due to changing efficiency competitiveness and evolutionary adaptations can all reasonably be expected. This could prove to be both beneficial and detrimental; ecological aspects have not been considered in this essay and certainty on how these developments will unfold requires further study. On a smaller and more concrete scale however, previously barren areas might be able to sustain crops after all, certain species might be applicable to agriculture where previously not deemed productive enough and deforestation might find a reducing factor in expanding growth potentials. Though in no way definitive, the survey here performed gives a cautiously optimistic view on global biosphere and agricultural developments.

7. Sources and notes

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7.4 Digital resources:

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http://www.uic.edu/classes/bios/bios100/labs/plantanatomy.htm

(image sources)

7.5 Notes

X). Due to the academic level of the essay and the author, some statements might be incorrect, outdated or unaware of parallel studies. No rights or liabilities may be derived from this.

8. Appendix A

Photosystem II:

$H_2O \rightarrow 2\ e^- + 2\ H^+ + 0.5\ O_2$ (light driven) P.1

Photosystem I:

$NADP^+ + H^+\ 2\ e^- \rightarrow NADPH$ (light driven, reversible to yield energy) P.2

Phosporylation:

$ADP + Pi \rightarrow ATP + H_2O$ (proton gradient driven, reversible to yield energy) P.3

Calvin cycle:

$3\ CO_2 + 9\ ATP + 6\ NADPH + 6\ H^+ \rightarrow PGAL + 9\ ADP + 8\ Pi + 6\ NADP^+ + 3\ H_2O$ C.1

$2\ PGAL + 2\ H_2O \rightarrow Glucose + 2\ Pi$ (and vice versa) C.3

Glycolysis:

$Glucose + 2\ NAD^+ + 2\ ADP + 2\ Pi \rightarrow 2\ Pyruvate + 2\ NADH + 2\ H^+ + 2\ ATP + 2\ H_2O$ R.1

$Pyruvate + NAD^+ + CoA \rightarrow CO_2 + Acetyl\ CoA + NADH + H^+$ R.2

Citric acid cycle:

$Oxaloacetate + Acetyl\ CoA + 3\ H_2O + ADP + Pi + 3\ NAD^+ + FAD \rightarrow Oxaloacetate + 2\ CO_2 + CoA + ATP + 3\ NADH + 3\ H^+ + FADH_2$ R.3

Carrier yield:

$3\ ADP + 3\ Pi + 2\ H^+ + O_2 + NADH \rightarrow 3\ ATP + 4\ H_2O + NAD^+$ R.4

$2\ ADP + 2\ Pi + 2\ H^+ + O_2 + FADH_2 \rightarrow 2\ ATP + 3\ H_2O + FAD$ R.5

$H_2O + NADP^+ + H^+ \rightarrow 2\ H^+ + 0.5\ O2 + NADPH$

$ADP + Pi \rightarrow ATP + H_2O$

$3\ CO_2 + 9\ ATP + 6\ NADPH + 6\ H^+ \rightarrow PGAL + 9\ ADP + 8\ Pi + 6\ NADP^+ + 3\ H_2O$

$PGAL + 2\ H_2O \rightarrow Glucose + 2\ Pi$

$Glucose + 2\ NAD^+ + 2\ ADP + 2\ Pi \rightarrow 2\ Pyruvate + 2\ NADH + 2\ H^+ + 2\ ATP + 2\ H_2O$

$Pyruvate + NAD^+ + CoA \rightarrow CO_2 + Acetyl\ CoA + NADH + H^+$

$Oxaloacetate + Acetyl\ CoA + 3\ H_2O + ADP + Pi + 3\ NAD^+ + FAD \rightarrow Oxaloacetate + 2\ CO_2 + CoA + ATP + 3\ NADH + 3\ H^+ + FADH_2$

$3\ ADP + 3\ Pi + 2\ H^+ + O_2 + NADH \rightarrow 3\ ATP + 4\ H_2O + NAD+$

$2\ ADP + 2\ Pi + 2\ H^+ + O_2 + FADH_2 \rightarrow 2\ ATP + 3\ H_2O + FAD$

$H_2O + NADP^+ + H^+ \rightarrow 2\ H^+ + 0.5\ O2 + NADPH$

$ADP + Pi \rightarrow ATP + H_2O$

$3\ CO_2 + 9\ ATP + 6\ NADPH + 6\ H^+ \rightarrow PGAL + 9\ ADP + 8\ Pi + 6\ NADP^+ + 3\ H_2O$

$2\ PGAL + 2\ H_2O \rightarrow Glucose + 2\ Pi$

Glucose + 10 NAD^+ + 4 ADP + 4 Pi + 4 H_2O + 2 FAD → 10 NADH + 10 H^+ + 4 ATP + 6 CO_2 + 2 $FADH_2$

3 ADP + 3 Pi + 2 H^+ + O_2 + NADH → 3 ATP + 4 H_2O + NAD^+
2 ADP + 2 Pi + 2 H^+ + O_2 + $FADH_2$ → 2 ATP + 3 H_2O + FAD

H_2O + $NADP^+$ + H^+ → 2 H^+ + 0.5 O_2 + NADPH

ADP + Pi → ATP + H_2O

6 CO_2 + 18 ATP + 12 NADPH + 12 H^+ → 18 ADP + 16 Pi + 12 $NADP^+$ + 2 H_2O + Glucose + 2 Pi

Glucose + 10 NAD^+ + 4 ADP + 4 Pi + 4 H_2O + 2 FAD → 10 NADH + 10 H^+ + 4 ATP + 6 CO_2 + 2 $FADH_2$

3 ADP + 3 Pi + 2 H^+ + O_2 + NADH → 3 ATP + 4 H_2O + NAD^+
2 ADP + 2 Pi + 2 H^+ + O_2 + $FADH_2$ → 2 ATP + 3 H_2O + FAD

H_2O + $NADP^+$ + H^+ → 2 H^+ + 0.5 O_2 + NADPH

ADP + Pi → ATP + H_2O

14 ATP + 12 NADPH + 2 H^+ + 10 NAD^+ + 2 FAD + 2 H_2O → 14 ADP + 14 Pi + 12 $NADP^+$ + 10 NADH + 2 $FADH_2$

3 ADP + 3 Pi + 2 H^+ + O_2 + NADH → 3 ATP + 4 H_2O + NAD^+
2 ADP + 2 Pi + 2 H^+ + O_2 + $FADH_2$ → 2 ATP + 3 H_2O + FAD

H_2O + $NADP^+$ + H^+ → 2 H^+ + 0.5 O_2 + NADPH

ADP + Pi → ATP + H_2O

10 ATP + 12 NADPH + 6 H^+ + 10 NAD^+ + 2 O_2 → 10 ADP + 10 Pi + 12 $NADP^+$ + 10 NADH + 4 H_2O

$3\ ADP + 3\ Pi + 2\ H^+ + O_2 + NADH \rightarrow 3\ ATP + 4\ H_2O + NAD^+$

$H_2O + NADP^+ + H^+ \rightarrow 2\ H^+ + 0.5\ O_2 + NADPH$

$ADP + Pi \rightarrow ATP + H_2O$

$12\ NADPH + 20\ ADP + 20\ Pi + 26\ H^+ + 12\ O_2 \rightarrow 12\ NADP^+ + 20\ ATP + 44\ H_2O$

$ADP + Pi \rightarrow ATP + H_2O$

$20\ ADP + 20\ Pi + 24\ H^+ + 6\ O_2 \rightarrow 20\ ATP + 32\ H_2O$

$ADP + Pi \rightarrow ATP + H_2O$

$24\ H^+ + 6\ O_2 \rightarrow 12\ H_2O$

$24H\ 12O \rightarrow 24H\ 12O$

Balanced

University of Amsterdam, 2009.